Extraction of Metals

Chemical Equations 17

Ores

Most metals are not found as shiny lumps in the ground – they are found in rocks called **ores**. Examples of ores are the rocks **hematite** (containing iron) and **malachite** (containing copper). The metal forms a **compound** in the ore; for example, the iron in hematite exists as the compound **iron oxide**. To extract the metal from its ore we need to separate the metal from its compound. We can achieve this separation using a **displacement reaction**.

Displacement Rule

A metal will be displaced from its ore by a **more reactive** element.

e.g. 1. Magnesium + Lead Oxide → Magnesium Oxide + Lead

Task 1

Q1. Why does the magnesium displace lead from its oxide?
Q2. Name 3 other metals in the reactivity series below that could displace lead from its oxide.

Extraction on a Larger Scale

The problem with using the above method to extract lead is that it is a waste of valuable magnesium. To extract metals on a larger scale, we need to use elements found in abundance in the earth's crust.

The reactivity series on the right includes carbon and hydrogen. Carbon is easily produced from coal. Hydrogen is produced from the gases found in the earth's crust. These two common elements can be used to extract metals from their ores.

Task 2

Write word equations for the following reactions. If you have been taught how to, then write a balanced symbol equation under each word equation (assume a valency of 3 for iron, 2 for zinc and tin, and 1 for copper).

	Reactants
a.	Carbon and iron oxide (producing carbon monoxide)
b.	Carbon and zinc oxide (producing carbon monoxide)
c.	Carbon and tin oxide (producing carbon monoxide)
d.	Carbon and copper oxide (producing carbon monoxide)
e.	Hydrogen and copper oxide

Task 3

List 5 metals that cannot be extracted from their oxides using carbon, and find out the name of the process used to extract these metals industrially.

Reactivity Series

most reactive

Potassium
Sodium
Calcium
Magnesium
Aluminum
Carbon
Zinc
Iron
Tin
Lead
Hydrogen
Copper

least reactive

Extraction of Iron on an Industrial Scale

The process of extracting iron on an industrial scale is a little more complicated. The extraction takes place in a large piece of equipment called a **blast furnace**.

What is added to the blast furnace?

a. Iron ore (containing a mixture of iron oxide and sand impurities)
b. Coke (pure carbon)
c. Limestone (calcium carbonate)
d. Air (containing oxygen)

What leaves the blast furnace?

a. Iron
b. Carbon dioxide
c. Slag (calcium silicate – $CaSiO_3$)

Task 4

For each process below, write a word equation to show the reaction taking place. Make sure you use the chemical name of each element or compound (i.e. use the name carbon instead of coke etc). If you have been taught how to, then write a balanced symbol equation under each word equation.

A. Extracting the iron

Process A1

The coke burns in the air to form carbon dioxide gas.

Process A2

This carbon dioxide reacts with more coke to produce carbon monoxide gas.

Process A3

This carbon monoxide reacts with the iron compound from the ore, producing molten iron and carbon dioxide gas. The molten iron is removed from the base of the blast furnace. The waste carbon dioxide is expelled through a vent.

B. Removing the sand

Process B1

The limestone decomposes in the heat to form quicklime (calcium oxide) and carbon dioxide gas.

Process B2

This quicklime fuses with the sand (silicon oxide) to produce slag. The slag is removed from the blast furnace.

Task 5

a. Which of the processes above is a Redox reaction, where one reactant is reduced (loses oxygen) and another is oxidized (gains oxygen)?

b. Which process is a displacement reaction?

c. Which process is an example of simple oxidation?

d. Which process is a thermal decomposition reaction, where a compound breaks down in the heat to form smaller components?